

Rural Detention

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Census API key has been set for this session.

1 RUCA Codes

Code	Classification description
1	Metropolitan core: primary flow is within an urban area of 50,000 or more people (metro UA)
2	Metropolitan high commuting: primary flow is 30 percent or more to a metro UA
3	Metropolitan low commuting: primary flow is 10 percent to 30 percent to a metro UA
4	Micropolitan core: primary flow is within an urban area of 10,000 to 49,999 people (micro UA)
5	Micropolitan high commuting: primary flow is 30 percent or more to a micro UA

Code	Classification description
6	Micropolitan low commuting: primary flow is 10 percent to 30 percent to a micro UA
7	Small town core: primary flow is within an urban area of 9,999 or fewer people (small town UA)
8	Small town high commuting: primary flow is 30 percent or more to a small town UA
9	Small town low commuting: primary flow is 10 percent to 30 percent to a small town UA
10	Rural area: primary flow is to a tract outside an UA
99	Not coded: census tract is entirely in coastal or inland water bodies, with zero population and zero land area

See the [RUCA Codes](#) documentation for detailed info on data and methods used for creating RUCA Codes.

	TractFIPS	TractName	CountyFIPS	CountyName	StateFIPS	StateName	RUCA
0	01001020100	Census Tract 201	01001	Autauga County	01	Alabama	1
1	01001020200	Census Tract 202	01001	Autauga County	01	Alabama	1
2	01001020300	Census Tract 203	01001	Autauga County	01	Alabama	1
3	01001020400	Census Tract 204	01001	Autauga County	01	Alabama	1
4	01001020501	Census Tract 205.01	01001	Autauga County	01	Alabama	1

	Count	Proportion
RUCA		
1	57544	68.69
2	8481	10.12
3	1149	1.37
4	4630	5.53
5	2224	2.65
6	693	0.83
7	1475	1.76
8	607	0.72
9	336	0.40
10	6343	7.57
99	294	0.35

2 ACS Data

2.1 Cost Burden

Collects variables for housing cost burden (renters and owners spending 30% or more of household income on housing) to measure overall cost burden.

Loaded cached variables for 2024 acs acs5

Loaded cached variables for 2024 acs acs5

	GEOID	CostBurden
0	01001020100	0.168478
1	01001020200	0.163851
2	01001020300	0.235628
3	01001020400	0.142857
4	01001020501	0.189303

2.2 Poverty Rate

Calculates poverty rate based on the number of people with income below the poverty level in 12 months and divides it by the population.

Loaded cached variables for 2024 acs acs5

	GEOID	PovertyRate
0	01001020100	0.174060
1	01001020200	0.095051
2	01001020300	0.071387
3	01001020400	0.153495
4	01001020501	0.035104

2.3 Housing Quality

Collects variables for substandard plumbing and kitchen facilities to measure overall housing quality.

Loaded cached variables for 2024 acs acs5

Loaded cached variables for 2024 acs acs5

	GEOID	HousingQuality
0	01001020100	0.007916
1	01001020200	0.100287
2	01001020300	0.092125
3	01001020400	0.008419
4	01001020501	0.000000

2.4 Education

Calculates percentage of the population that does not hold a high school diploma

Loaded cached variables for 2024 acs acs5

	GEOID	LessThanHS
0	01001020100	0.094197
1	01001020200	0.097473
2	01001020300	0.130983
3	01001020400	0.053578
4	01001020501	0.031766

2.5 Demographics

Loaded cached variables for 2024 acs acs5

	GEOID	PercentMinority
0	01001020100	0.334643
1	01001020200	0.676971
2	01001020300	0.358540
3	01001020400	0.119048
4	01001020501	0.215122

2.6 Household Vars

	GEOID	Gini
0	01001020100	0.4238
1	01001020200	0.4100
2	01001020300	0.3519
3	01001020400	0.5880

	GEOID	Gini
4	01001020501	0.3954

3 Data Analysis

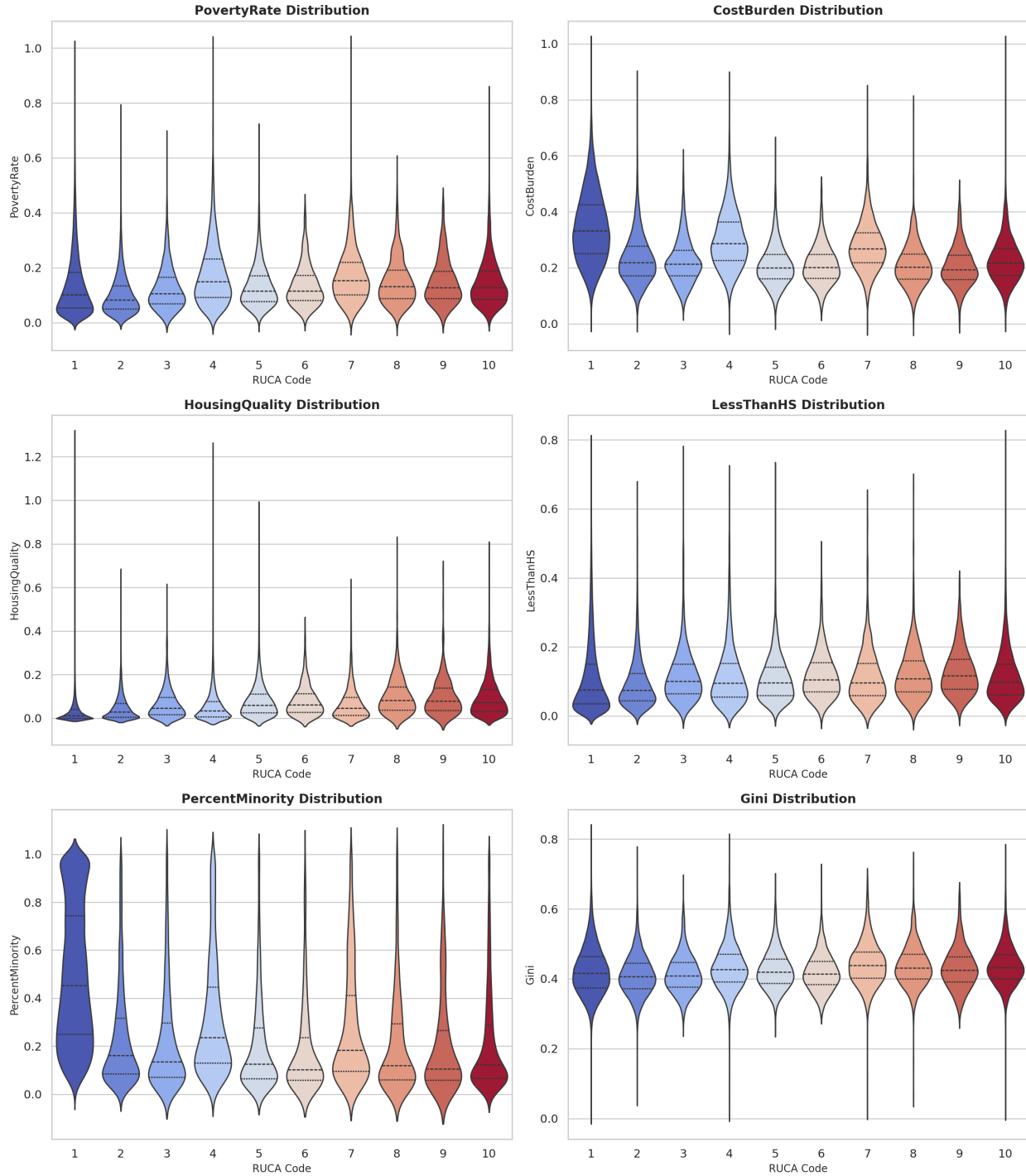
Merge all individual ACS socioeconomic indicators and combines them with the cleaned RUCA dataset by Census Tract GEOID.

	TractFIPS	TractName	CountyFIPS	CountyName	StateFIPS	StateName	RUCA	GEOID
0	01001020100	Census Tract 201	01001	Autauga County	01	Alabama	1	01001020100
1	01001020200	Census Tract 202	01001	Autauga County	01	Alabama	1	01001020200
2	01001020300	Census Tract 203	01001	Autauga County	01	Alabama	1	01001020300
3	01001020400	Census Tract 204	01001	Autauga County	01	Alabama	1	01001020400
4	01001020501	Census Tract 205.01	01001	Autauga County	01	Alabama	1	01001020501

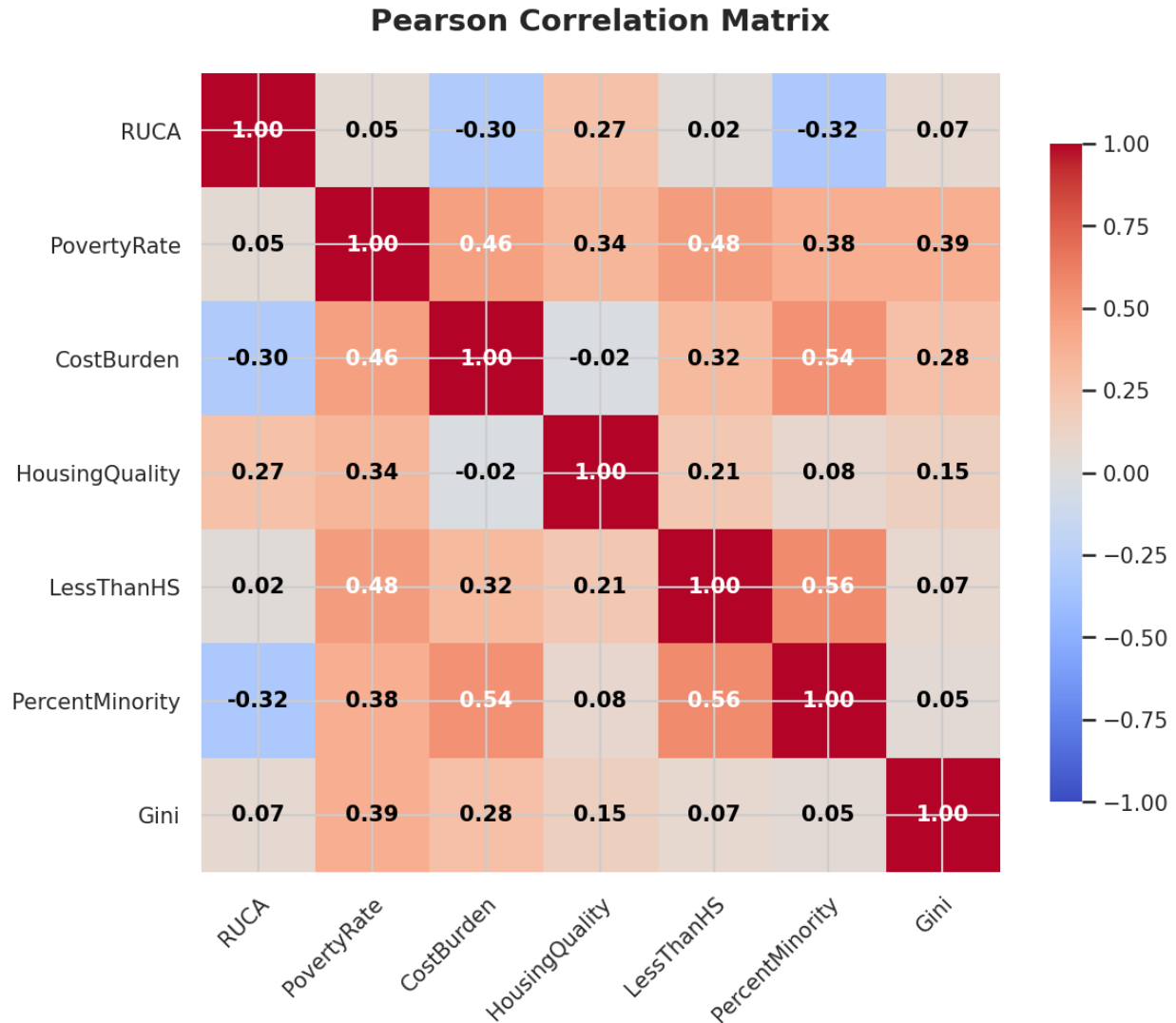
(82879, 81802)

3.1 Descriptive Statistics

	RUCA	PovertyRate	CostBurden	HousingQuality	LessThanHS	PercentMinority	Gini
count	81802.000000	81802.000000	81802.000000	81802.000000	81802.000000	81802.000000	81802.000000
mean	2.314088	0.135813	0.312121	0.044292	0.109586	0.416713	0.423781
std	2.628321	0.108560	0.125279	0.069683	0.096994	0.291475	0.067424
min	1.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000100
25%	1.000000	0.058684	0.217634	0.000000	0.041479	0.165269	0.378300
50%	1.000000	0.106323	0.294195	0.020309	0.081125	0.346462	0.417900
75%	2.000000	0.180921	0.391065	0.058467	0.147231	0.640236	0.463000
max	10.000000	1.000000	1.000000	1.307692	0.800313	1.000000	0.826000



3.2 Pearson Correlation Matrix



- The minority percentage and cost burden percentage of a census tract decreases as census tracts become more urban whereas poor housing quality goes up.
- LessThanHS has decent positive correlations (0.3 - 0.5) with PovertyRate and PercentMinority

3.3 Variable Inflation Factor

feature		VIF
0	PovertyRate	4.343494
1	CostBurden	12.310752
2	HousingQuality	1.680772
3	LessThanHS	3.834566
4	PercentMinority	5.660608

	feature	VIF
5	Gini	8.526785

- [VIF Interpretation](#)

- VIF = 1: No correlation with other predictors
- $1 < \text{VIF} \leq 5$: Mild to moderate correlation (usually fine)
- VIF > 10: Strong multicollinearity -> take corrective steps

CostBurden and Percent Minority are over 5 but less than 6, so I don't think they'll cause significant multicollinearity issues.

At 8.5 the Gini coefficient is close to being problematic. Since it's the main way the ACS measures income inequality, so unless it causes problems down the road I guess it's fine $\backslash()/$

3.4 Multinomial Logistic Regression

Optimization terminated successfully.

Current function value: 0.931436

Iterations: 375

Function evaluations: 377

Gradient evaluations: 377

MNLogit Regression Results

```
=====
Dep. Variable:          RUCA    No. Observations:          81802
Model:                  MNLogit  Df Residuals:              81739
Method:                  MLE     Df Model:                54
Date:                   Sun, 07 Jun 2026  Pseudo R-squ.:         0.2071
Time:                   13:39:49    Log-Likelihood:        -76193.
converged:               True      LL-Null:              -96090.
Covariance Type:        nonrobust  LLR p-value:           0.000
=====
```

	RUCA=2	coef	std err	z	P> z	[0.025	0.975]
const		0.7639	0.091	8.381	0.000	0.585	0.943
PovertyRate		-0.6955	0.220	-3.164	0.002	-1.126	-0.265
CostBurden		-8.1085	0.171	-47.294	0.000	-8.445	-7.772
HousingQuality		6.3021	0.232	27.109	0.000	5.846	6.758
LessThanHS		9.4826	0.214	44.409	0.000	9.064	9.901
PercentMinority		-5.0494	0.081	-62.492	0.000	-5.208	-4.891
Gini		0.6074	0.231	2.632	0.008	0.155	1.060

	RUCA=3	coef	std err	z	P> z	[0.025	0.975]
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const	-1.3342	0.224	-5.956	0.000	-1.773	-0.895
PovertyRate	2.3709	0.472	5.024	0.000	1.446	3.296
CostBurden	-10.0531	0.426	-23.574	0.000	-10.889	-9.217
HousingQuality	8.2449	0.433	19.042	0.000	7.396	9.094
LessThanHS	13.2341	0.437	30.263	0.000	12.377	14.091
PercentMinority	-6.8611	0.193	-35.506	0.000	-7.240	-6.482
Gini	0.8517	0.563	1.512	0.130	-0.252	1.955

	RUCA=4	coef	std err	z	P> z	[0.025	0.975]
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const	-1.6676	0.109	-15.243	0.000	-1.882	-1.453
PovertyRate	5.4197	0.177	30.537	0.000	5.072	5.768
CostBurden	-3.9845	0.179	-22.309	0.000	-4.335	-3.634
HousingQuality	4.9088	0.239	20.550	0.000	4.441	5.377
LessThanHS	6.5278	0.224	29.102	0.000	6.088	6.967
PercentMinority	-4.5614	0.089	-51.123	0.000	-4.736	-4.387
Gini	1.1015	0.266	4.145	0.000	0.581	1.622

	RUCA=5	coef	std err	z	P> z	[0.025	0.975]
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const	-1.4349	0.164	-8.726	0.000	-1.757	-1.113
PovertyRate	4.2239	0.340	12.406	0.000	3.557	4.891
CostBurden	-13.3378	0.333	-40.054	0.000	-13.991	-12.685
HousingQuality	9.7721	0.299	32.669	0.000	9.186	10.358
LessThanHS	11.3313	0.368	30.781	0.000	10.610	12.053
PercentMinority	-6.9921	0.146	-47.879	0.000	-7.278	-6.706
Gini	4.1201	0.405	10.172	0.000	3.326	4.914

	RUCA=6	coef	std err	z	P> z	[0.025	0.975]
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const	-1.9850	0.286	-6.939	0.000	-2.546	-1.424
PovertyRate	4.5077	0.558	8.073	0.000	3.413	5.602
CostBurden	-12.5656	0.572	-21.986	0.000	-13.686	-11.445
HousingQuality	9.6827	0.489	19.793	0.000	8.724	10.642
LessThanHS	13.7956	0.554	24.892	0.000	12.709	14.882
PercentMinority	-8.1782	0.256	-31.979	0.000	-8.679	-7.677
Gini	2.0686	0.710	2.913	0.004	0.677	3.460

	RUCA=7	coef	std err	z	P> z	[0.025	0.975]
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const	-3.5833	0.184	-19.433	0.000	-3.945	-3.222
PovertyRate	5.5806	0.306	18.224	0.000	4.980	6.181
CostBurden	-6.2996	0.317	-19.884	0.000	-6.921	-5.679

HousingQuality	6.8768	0.349	19.714	0.000	6.193	7.560
LessThanHS	7.8038	0.396	19.699	0.000	7.027	8.580
PercentMinority	-5.5173	0.155	-35.610	0.000	-5.821	-5.214
Gini	4.3665	0.442	9.884	0.000	3.501	5.232

	RUCA=8	coef	std err	z	P> z	[0.025 0.975]

const	-3.8594	0.295	-13.087	0.000	-4.437	-3.281
PovertyRate	5.0861	0.562	9.053	0.000	3.985	6.187
CostBurden	-14.1110	0.590	-23.910	0.000	-15.268	-12.954
HousingQuality	11.2457	0.431	26.065	0.000	10.400	12.091
LessThanHS	12.7613	0.604	21.123	0.000	11.577	13.945
PercentMinority	-7.8149	0.251	-31.090	0.000	-8.308	-7.322
Gini	6.5252	0.706	9.248	0.000	5.142	7.908

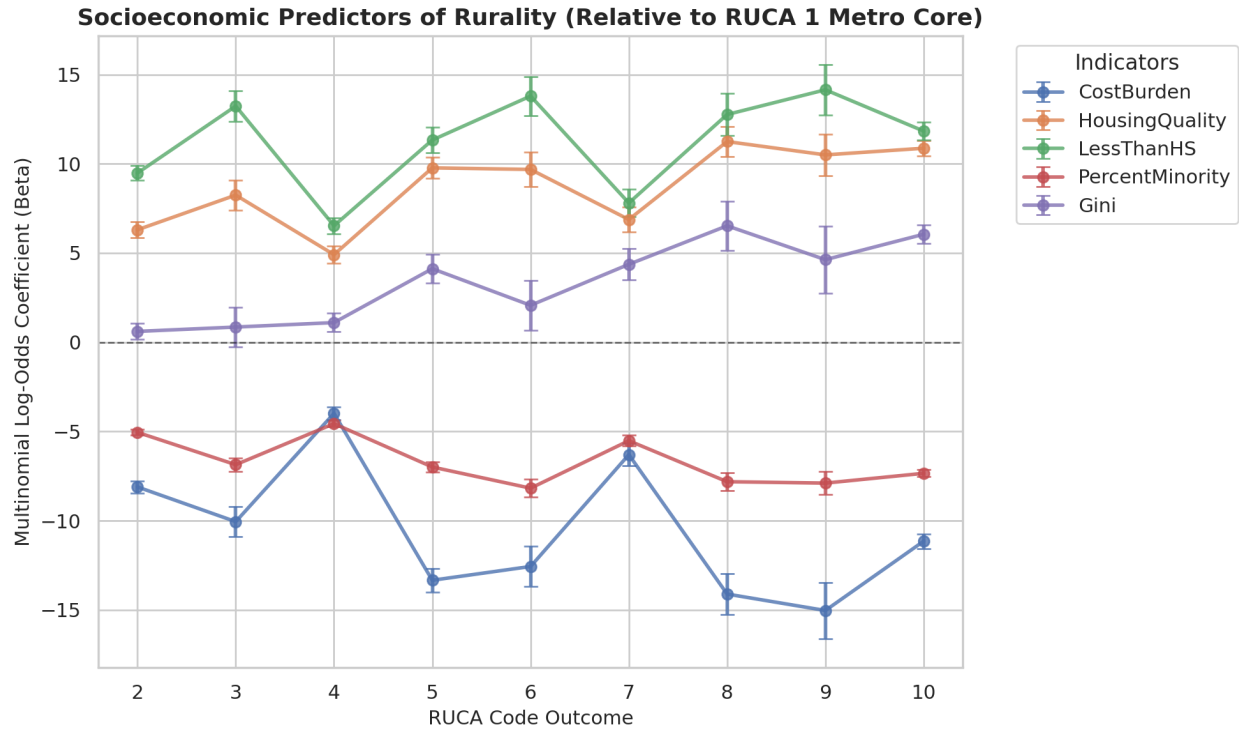
	RUCA=9	coef	std err	z	P> z	[0.025 0.975]

const	-3.5497	0.392	-9.046	0.000	-4.319	-2.781
PovertyRate	5.2732	0.720	7.324	0.000	3.862	6.684
CostBurden	-15.0318	0.801	-18.757	0.000	-16.603	-13.461
HousingQuality	10.4987	0.594	17.660	0.000	9.334	11.664
LessThanHS	14.1473	0.726	19.473	0.000	12.723	15.571
PercentMinority	-7.8935	0.327	-24.111	0.000	-8.535	-7.252
Gini	4.6274	0.953	4.855	0.000	2.759	6.495

	RUCA=10	coef	std err	z	P> z	[0.025 0.975]

const	-1.7874	0.110	-16.211	0.000	-2.003	-1.571
PovertyRate	4.0689	0.222	18.322	0.000	3.634	4.504
CostBurden	-11.1487	0.212	-52.605	0.000	-11.564	-10.733
HousingQuality	10.8718	0.214	50.712	0.000	10.452	11.292
LessThanHS	11.8330	0.249	47.512	0.000	11.345	12.321
PercentMinority	-7.3449	0.100	-73.477	0.000	-7.541	-7.149
Gini	6.0515	0.269	22.535	0.000	5.525	6.578
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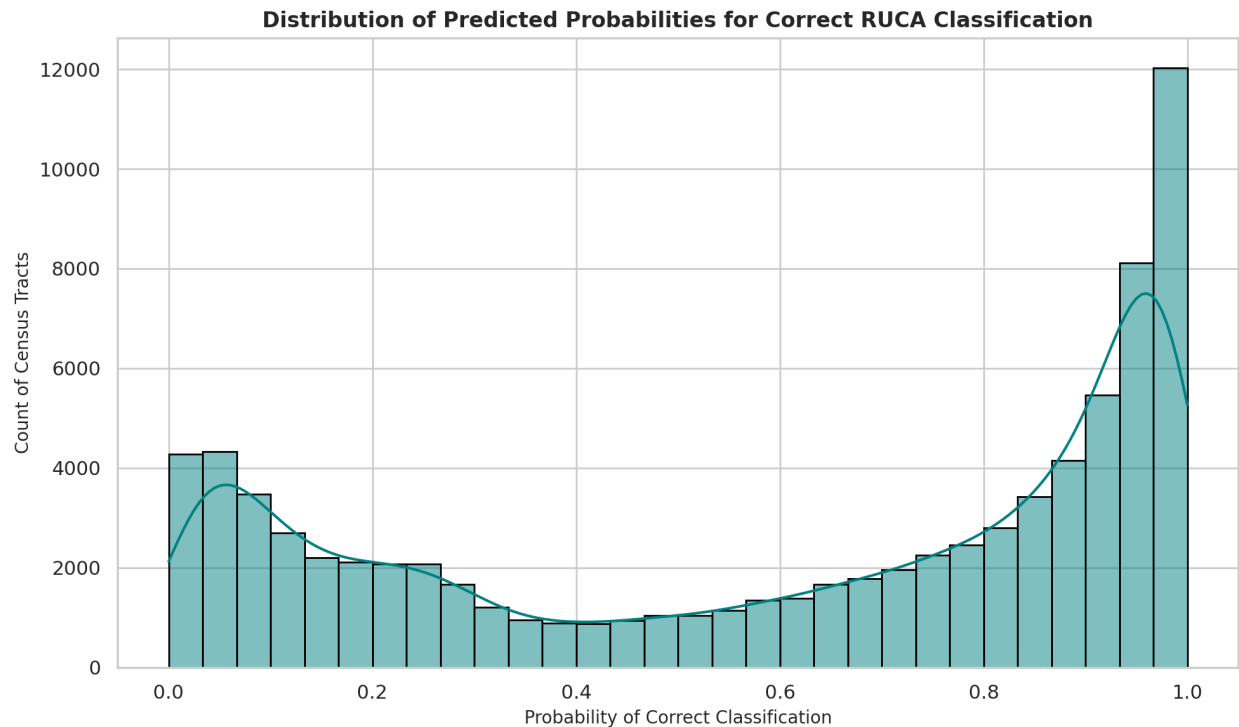
Pretty much all of the coefficients are statistically significant but considering the large number of observations and the very low amount of variation between values, I don't think it really means much beyond being able to say they're statistically significant.



Positive multinomial log-odds coefficient: variable increase -> likelihood of class (RUCA code) increases.

Mean probability of correct classification: 59.81

Median probability of correct classification: 0.7298630414056722



It's certainly one of the distributions ever made.

4 Spatial Analysis